

ALGONA, IA, USA

WMO: 725457

Lat: 43.078N

Lon: 94.272W

Elev: 1219

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-4.2	-16.7	2.3	-8.2	-11.3	3.1	-4.1	32.9	15.5	29.0	16.1	9.5	320	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.6	76.1	88.2	74.9	85.6	73.2	79.8	86.8	77.6	84.3	75.3	82.7	14.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	149.1	84.5	75.1	137.8	82.7	72.8	127.4	80.2	44.3	87.2	41.9	84.7	39.7	82.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	25.0	22.3	-18.1	94.4	5.1	3.2	-21.7	96.7	-24.7	98.6	-27.6	100.3	-31.2	102.6		
(5)			-16.0	83.1	7.1	2.3	-21.1	84.7	-25.2	86.0	-29.2	87.3	-34.4	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	18.4	22.4	34.3	47.3	59.8	70.2	73.1	70.6	64.0	50.2	36.3	23.6		
	DBStd	21.71	12.40	12.35	12.46	10.94	8.90	6.52	5.69	5.52	8.06	9.64	11.16	11.57		
	HDD50	3822	980	772	507	176	17	0	0	0	4	117	429	820		
	HDD65	7110	1445	1192	952	535	211	26	6	14	115	468	862	1284		
	CDD50	2963	0	0	21	96	322	605	717	639	423	123	17	0		
	CDD65	778	0	0	1	6	50	181	258	188	84	10	0	0		
	CDH74	7035	0	0	8	117	525	1655	2369	1463	770	124	4	0		
	CDH80	2242	0	0	1	32	168	577	807	408	212	37	0	0		
	WSAvg	10.6	11.6	11.8	12.1	12.8	12.0	9.9	7.8	7.1	8.8	10.8	11.5	11.2		
	PrecAvg	30.5	0.7	0.8	1.8	3.0	3.9	4.6	4.2	3.7	3.2	2.1	1.5	0.9		
(15) Precipitation	PrecMax	46.0	2.4	3.5	5.2	7.2	8.0	11.0	9.0	9.1	11.5	5.5	5.6	3.4		
	PrecMin	17.2	0.0	0.0	0.1	0.7	1.5	1.3	0.9	0.3	0.6	0.3	0.0	0.0		
	PrecStd	6.7	0.5	0.7	1.3	1.6	1.7	2.4	2.0	2.3	2.3	1.4	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	54.9	72.8	84.2	91.0	93.4	93.4	90.8	90.3	85.3	71.8	55.5		
		MCWB	41.9	51.0	58.5	64.6	71.0	73.8	80.1	78.5	72.4	65.6	58.5	49.3		
	2%	DB	40.7	46.1	64.0	76.8	84.0	90.2	90.5	87.9	86.0	76.7	63.4	47.6		
		MCWB	37.1	42.6	55.7	59.4	66.8	73.7	78.9	77.1	71.6	60.4	54.5	44.2		
	5%	DB	36.5	41.2	58.6	71.6	79.5	86.2	88.0	84.3	81.7	71.7	57.1	41.3		
		MCWB	34.4	38.0	52.2	57.0	64.3	71.2	77.5	74.5	69.2	59.3	49.5	37.8		
	10%	DB	34.1	37.1	52.1	64.5	75.0	82.3	84.3	81.6	78.9	65.9	52.0	37.0		
		MCWB	32.5	34.8	47.5	53.3	61.7	69.2	75.1	72.3	66.9	56.4	46.1	34.7		
	0.4%	WB	42.3	50.7	63.1	66.7	73.6	79.8	83.7	81.2	76.7	68.8	60.8	52.2		
		MCDB	47.7	55.3	71.1	80.3	84.7	87.7	91.0	88.3	84.0	79.9	68.5	55.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.4	43.1	57.1	62.0	69.6	76.4	80.8	78.3	74.3	65.1	55.3	43.3		
		MCDB	40.0	46.3	62.7	72.2	80.7	85.3	86.7	84.9	82.0	71.9	62.0	46.3		
	5%	WB	34.6	38.5	52.3	58.7	67.0	74.3	78.8	75.9	72.1	60.9	50.2	38.5		
		MCDB	36.4	41.3	58.2	69.0	76.0	82.9	85.6	82.5	79.2	68.3	56.3	41.4		
	10%	WB	32.3	35.2	46.6	54.8	64.2	72.0	76.5	73.8	69.6	56.9	45.9	35.2		
		MCDB	33.6	37.4	51.6	63.5	72.0	80.5	83.3	80.1	75.7	64.5	50.9	37.5		
	5% DB	MDBR	15.3	15.5	16.0	20.2	19.8	19.5	19.0	18.6	21.8	20.5	17.3	14.1		
		MCDBR	17.6	17.7	25.5	30.9	25.6	23.4	20.4	20.5	24.6	27.7	25.9	19.3		
		MCWBR	15.1	15.0	17.1	17.5	12.9	11.2	12.3	12.4	13.2	15.3	16.8	15.7		
		MCWBR	16.8	17.3	24.0	27.3	22.3	19.8	18.9	18.5	19.9	23.1	24.5	17.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.6	14.8	17.0	17.3	12.6	12.2	12.9	12.8	12.3	15.2	17.0	14.9		
	Clear-Sky Solar Irradiance	taub	0.283	0.284	0.312	0.364	0.388	0.412	0.421	0.409	0.373	0.333	0.291	0.268		
		taud	2.406	2.419	2.447	2.330	2.334	2.313	2.264	2.291	2.357	2.462	2.511	2.476		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	272	293	297	286	281	273	269	269	271	270	269	270		
		Edh at Noon	24	28	31	38	39	40	42	39	34	27	22	21		
(44) All-Sky Solar Radiation	RadStd		587	919	1203	1493	1758	1947	2046	1720	1393	928	622	461		
			62	90	121	141	126	153	101	121	93	120	48	36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page